

**Task Name:****Multi-Channel Marketing Attribution with High-Dimensional Markov Chain Transitions**

## 1. Introduction

Marketing is done through different channels such as social media, email, search engines, advertisements and websites. A customer may interact with many channels before finally purchasing a product. Therefore, it is difficult to identify which channel is mainly responsible for the conversion.

The **Markov Chain model** helps to study these customer journeys and find the contribution of each marketing channel.

## 2. Data Used

In this task, **15 million user path logs** are used. Each log represents the sequence of marketing channels visited by a customer.

**Example:**

Social Media → Website → Email → Search → Conversion

These paths are used to understand how customers move between different channels.

## 3. Transition Probability Matrix

A transition probability matrix is created from the user paths.

It shows the probability of moving from one channel to another.

For example:

| From/To      | Social Media | Email | Searching | Conversion |
|--------------|--------------|-------|-----------|------------|
| Social Media | 0.20         | 0.30  | 0.30      | 0.20       |
| Email        | 0.10         | 0.40  | 0.40      | 0.30       |
| Search       | 0.15         | 0.20  | 0.20      | 0.40       |

## 4. Conversion Probability

Using the transition matrix, the probability that a customer finally reaches the **Conversion** state is calculated.

This is taken as the **original conversion probability**.

## 5. Removal Effect

To find the importance of each marketing channel, one channel is removed from the transition model at a time.

For example, first remove **Email** and calculate the new conversion probability.

Then remove **Social Media**, **Search**, etc., one by one.

The formula is:

$$\text{Removal Effect} = \frac{P_{\text{original}} - P_{\text{removed}}}{P_{\text{original}}}$$

Where:

- $P_{\text{original}}$  = conversion probability before removing the channel
- $P_{\text{removed}}$  = conversion probability after removing the channel

## 6. Example Result

Assume:

**Original conversion probability = 0.40**

After removing Email:

**New conversion probability = 0.30**

Therefore,

$$\text{Removal Effect} = \frac{0.40 - 0.30}{0.40} = 0.25 = 25\%$$

So, **Email has a 25% Removal Effect** in this example

## 7. Final Result

After calculating the Removal Effect for all channels, the channels can be ranked based on their contribution.

| Marketing Channel | Removal Effect |
|-------------------|----------------|
| Email             | 25%            |
| Social Media      | 20%            |

|               |     |
|---------------|-----|
| Search        | 35% |
| Advertisement | 15% |

**Note:** These values are only an example. Actual values should be calculated from the 15 million user logs.

## 8. Conclusion

The Markov Chain model provides a systematic way to measure the contribution of different marketing channels. The Removal Effect shows how conversion probability changes when a channel is removed. Therefore, businesses can identify important channels and use their marketing budget more effectively.